

ME302 : Energy System II Course Project

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1 Problem Description

The objective of this project is to determine the maximum feasible geometric scale of a compressor test model while ensuring dynamic similarity with a given prototype and satisfying operational constraints of a closed-loop experimental facility.

The prototype operates under specified inlet conditions, and the model must replicate its aerodynamic behavior by maintaining similarity in terms of Reynolds number and Mach number. The working fluid is air, and the test section conditions are constrained by a minimum allowable static pressure.

The scale factor we use here S is defined as:

$$S = \frac{l_{\text{model}}}{l_{\text{prototype}}} \quad (1)$$

The model must satisfy the following conditions:

1.1 Reynolds Number Similarity

The flow in the model must match the prototype Reynolds number:

$$Re = \frac{\rho V l}{\mu} \quad (2)$$

where ρ is density, V is velocity, l is characteristic length for model, and μ is dynamic viscosity.

1.2 Mach Number Similarity

The Mach number at the test section is fixed:

$$M = 0.5 \quad (3)$$

1.3 Minimum Static Pressure Constraint

The static pressure in the test section must satisfy:

$$p_4 \geq 200 \text{ kPa} \quad (4)$$

1.4 Compressor Operating Constraints

The compressor must operate within the provided compressor map, which defines the stagnation pressure ratio as:

$$PR = \frac{p_{02}}{p_{01}} \quad (5)$$

2 METHODOLOGY

2.1 Derivations

Temperature at station 4. The facility is adiabatic, so stagnation temperature is constant beyond station 2: $T_{04} = T_{02} = (T_{02}/T_{01}) \times T_{01}$. We can use the relation between stagnation and static temperature at point 4 as :

$$T_4 = T_{04} / (1 + \frac{\gamma - 1}{2} \times Ma^2) = T_{04} / 1.05 \quad (6)$$

Mass flow rate Using $Re = \rho_4 C_4 l_m / \mu$, $A_4 = \pi l_m^2$ (since $D_4 = 2l_m$), and $l_m = l_p \times s$:

$$\dot{m} = \rho_4 C_4 A_4 = Re \times \mu \times \pi \times l_p \times s \quad (7)$$

Pressure at station 4. From ideal gas law and isentropic stagnation:

$$p_{04} = \rho_4 R T_4 \times (T_{04}/T_4)^{\gamma/(\gamma-1)} = \left[\frac{Re \mu \sqrt{RT_4}}{Ma \sqrt{\gamma} \times l_p \times s} \right] \times (1.05)^{3.5} \quad (8)$$

Pressure chain. For $s \leq 1$, $D_4 = 2 \times 0.2 \times s \leq 0.4 \text{ m} < D_3 = 0.6 \text{ m}$, so the section 3→4 is always a converging nozzle ($\theta < 0$), so that $K = 0$ and $p_{03} = p_{04}$. The exit stagnation pressure is at greater than or equal to the inlet stagnation pressure:

$$p_{05} \geq p_{01} \quad (9)$$

Accounting for pressure losses in the system:

$$p_{03} = 0.985 p_{02} \quad (10)$$

$$p_{04} = p_{03} \quad (11)$$

$$p_{05} = 0.95 p_{04} \quad (12)$$

$$p_{03} = p_{04} \rightarrow p_{02} = p_{03} / 0.985 \rightarrow p_{01} = p_{02} / (p_{02}/p_{01}) \rightarrow p_{05} = 0.95 \times p_{04} \quad (13)$$

The mass flow rate through the model must be consistent with the compressor map values, adjusted for operating pressure for $T_{01} = 293$ we have :

$$\dot{m}_{\text{model}} = \dot{m}_{\text{ref}} \left(\frac{p_{01}}{p_{\text{ref}}} \right) \quad (14)$$

Actual mass flow rate must equal the compressor map value:

$$\dot{m}_{\text{ref}} \times (p_{01} / 101.325 \text{ kPa}) = Re \times \mu \times \pi \times l_p \times s \quad (15)$$

Then we put the value of P_{01} using all pressure relations and we get :

$$s^2 = \frac{\dot{m}_{\text{ref}} \times \sqrt{RT_4} \times (1.05)^{3.5}}{p_{\text{ref}} \times \pi \times l_p^2 \times Ma \times \sqrt{\gamma} \times 0.985 \times (p_{02}/p_{01})} \quad (16)$$

Analysis: Analysis was performed in MATLAB 3.

3 RESULTS AND DISCUSSION

3.1 Constraint Analysis

All tabulated compressor operating points were found to satisfy $p_4 \geq 200$ kPa and $p_{05} \geq p_{01}$. The binding constraint is therefore the scale ceiling $s \leq 1$ together with maximizing s using relation obtained in equation with $\dot{m}_{ref}$, PR and TR.

3.2 Optimal Solution

The table below summarizes the results at the optimal (maximum-scale) operating condition (Pressure and temp rounded off to 2 decimal places):

Parameter	Value	Unit
Maximum scale, $s = l_{model}/l_{prototype}$	0.638449	—
Model length, l_{model}	12.77	cm
Test section inlet diameter, $D_4 = 2 \times l_{model}$	25.54	cm
Diffuser semi-angle, θ	-6.55	° (nozzle)
Inlet stagnation pressure, p_{01}	211.51	kPa
Compressor $\dot{m}_{ref}$	10.55028	kg/s
Compressor T_0 ratio (T_{02}/T_{01})	1.068519	—
Compressor p_0 ratio (p_{02}/p_{01})	1.210454	—
Stagnation temperature $T_{02} = T_{04}$	313.08	K
Static temperature at station 4, T_4	298.17	K
Velocity at station 4, C_4	173.06	m/s
Static pressure at station 4, p_4	212.60	kPa
Outlet stagnation pressure, p_{05}	239.57	kPa
Actual mass flow rate, $\dot{m}$	22.02	kg/s
Reynolds number (verification)	3.00×10^6	—

Table 1: Results at the optimal operating condition.

3.3 Compressor Operating Point

The following figure shows the compressor characteristic curve with the selected operating point marked. The point lies at the high-flow, low-pressure-ratio end of the map, which maximizes s by requiring a lower p_{01} , thus relaxing the $p_4 \geq 200$ kPa constraint.

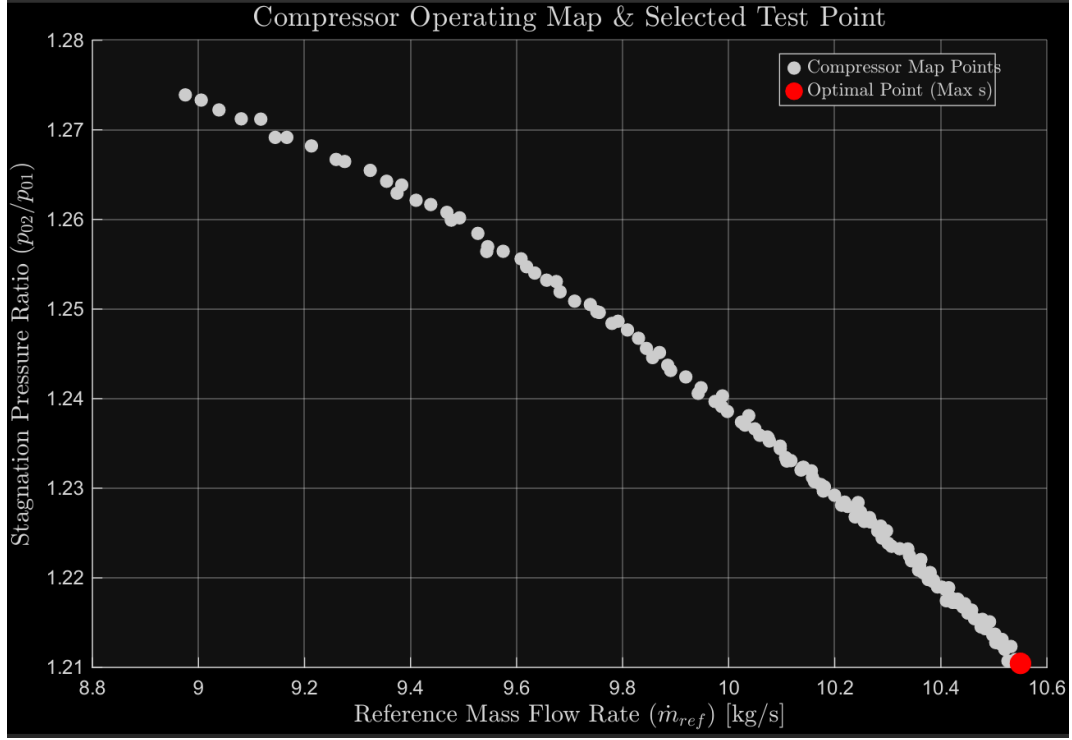


Figure 1: Compressor operation map with the selected maximum-scale operating point.

3.4 Verification of Constraints

1. **Minimum pressure constraint:** The lowest static pressure occurs at station 4. $p_4 = 212.60$ kPa ≥ 200 kPa.
2. **Closed-loop constraint:** $p_{05} = 239.57$ kPa $\geq p_{01} = 211.51$ kPa.
3. **Mach and Reynolds Matching:** $Ma_4 = 0.5$ and $Re = 3.00 \times 10^6$ are satisfied by construction.

4 SUPPLEMENT

The following files are submitted along with this report:

- ME302.m — MATLAB script for calculations and plot generation.
- Compressor_operation_map.xlsx — Provided data file.
- compressor_map.png — Generated compressor map plot.